BOOK II

Osnias Clearing Network Architecture

Release 3.0 — Working Draft | 26 October 2026

Author: Denis Bouzon | ORCID 0009-0007-8894-8902

Reference page: https://www.osnias-clearing.com/architecture/

Status: proposed technical architecture; unpublished, unaudited and subject to banking, contractual and regulatory validation.

Executive Summary

Osnias Clearing is a software infrastructure for the multilateral clearing of B2B trade obligations. The canonical ledger, developed in standard Solidity and portable across EVM-compatible chains, is deployed on Sei EVM as its reference environment. Cover funds are held as commercial bank money in individual escrow accounts, one per Osnias-ID, company, currency and cycle profile. The bank attests that the funds are effectively locked. A shared Bank Escrow Oracle queries the bank on demand, validates the escrow status and transmits verifiable evidence to the Core. The ledger neither holds, issues nor transfers clearing tokens or stablecoins. Net balances are settled by banks or authorised service providers pursuant to the applicable agreements. Solvency remains subject to a non-negative position for every participant.

Keywords: B2B clearing, multilateral netting, Solidity, Sei EVM, portable architecture, bank escrow, IBAN, bank attestation, oracle, ISO 20022, Osnias-ID, net settlement.

1. Document Position and Status

This Book II describes the target functional architecture. It supersedes Release 2.1 of 14 October 2026, which described tokenised collateral on Ethereum, cross-chain Controllers and Ethereum/Sei reconciliation checks. Release 3.0 replaces those components with bank escrow. Books I, III, IV, V, VI and VII must be aligned before normative publication; their published editions are not amended by this working draft.

2. Architectural Principles

Osnias Core executes the rules for obligation validation, position updates and multilateral netting on a single canonical chain, Sei EVM. Business logic is written in standard EVM Solidity without Sei-specific libraries, precompiles or instructions. Configurable adapters handle network-specific dependencies. No bank-escrow smart contract is required.

Custody of funds, enforcement of their unavailability and settlement transfers are entrusted to banks. The banking oracle operates solely as a read, verification and evidence interface: it does not create escrow and has no authority to initiate transfers.

3. Clearing, Discharge of Obligations and Settlement

3.a Clearing

An obligation request is identified and submitted for acceptance. Its admission updates the positions of the Osnias-ID accounts without transferring bank money. The calculation produces net balances by member, currency and clearing chamber. Clearing does not automatically discharge the underlying claim: any discharge effect is governed by the rulebook, agreements and applicable law.

3.b Bank Settlement

The bank or authorised provider executes net settlement instructions. Osnias Core retains references and reconciliation statuses without being able to debit accounts. Legal finality, settlement incidents, disputes and insolvency are governed contractually; no automatic protection under Directive 98/26/EC is presumed.

The net positions established by the Core give rise to instructions individualised by escrow account and account-servicing bank. Duly mandated node gateways transmit these instructions to the banks in ISO 20022 format through agreed channels. Only the bank may release the lock within the authorised scope in order to debit the payable net amount and execute settlement. A bank message acknowledging receipt of an order does not establish that settlement has become final.

4. Functional Architecture and Separation of Roles

Layer	Function	Limit of Responsibility
Osnias Core — Sei EVM	Obligations ledger, positions, cycles, admission checks	No custody or payment execution
Bank Escrow Oracle	Bank queries, evidence validation and authenticated attestation delivery	No fund release or transfer
Banking institution	Dedicated IBAN, locking of funds, statements and settlement	Contractually defined obligations
Node operator	Onboarding, identity, ERP, compliance and client relationship	Crypto authorisation not presumed necessary; qualification to be assessed
ERP	Issuance, reconciliation and accounting of obligations	No solvency decision

4.a Federated Architecture

Nodes remain independent and operate within their respective jurisdictions. They administer client access to the protocol and ERP interfaces. The banking infrastructure may comprise multiple banks. Escrow verification rules are common to the entire network, while each adapter ensures compatibility with the APIs of a particular bank.

4.b Dedicated Escrow Account

Each active Osnias-ID corresponds to an individual bank escrow account with a distinct IBAN, assigned to one company, one ISO 4217 currency and one cycle profile (1W, 9W or 13W). Funds are not pooled between companies. The legal and technical escrow status arises from the banking agreement and restrictions actually enforced by the bank. A positive balance or a revocable account freeze alone is not sufficient evidence.

5. Osnias-ID and Secure Banking Registry

The Osnias-ID retains its normative format defined in Book I (19 digits, MOD-97 check digits). The Osnias-ID ↔️ IBAN ↔️ bank ↔️ account holder ↔️ currency ↔️ cycle mapping registry is off-chain, protected and accessible only to authorised parties. Neither the IBAN, the legal identity nor invoice references may be published in cleartext on Sei. The Core retains only pseudonymous identifiers and evidence hashes required for control and audit.

6. Escrow Operational Cycle

6.a Funding and Activation

Before the active period opens, the bank verifies the existence of funds, locks them irrevocably for the agreed period and attests the amount, date, IBAN, duration, debit restrictions and release rights. Osnias Core activates capacity C_i only after validating genuine, current bank evidence linked to the correct Osnias-ID and cycle.

6.b Clearing Period

During the active cycle, the cover funds remain in escrow. The account holder cannot make ordinary withdrawals or transfers from the locked amount. The oracle may query the bank at any time at the Core's request, subject to a contractually accessible API and a data-freshness commitment. An outage or stale response prevents admission of any new obligation requiring that cover; it does not cancel previously accepted obligations.

6.c Closing, Disputes, Settlement and Release

At closing, the Core freezes the clearing state and produces a deterministic, reconcilable settlement batch identified by chamber, currency, cycle, Osnias-ID and unique reference. At most one net debit instruction is generated for each debtor account in that batch, subject to exceptions and settlement procedures. Creditor accounts receive the corresponding net credits through the banks; interbank transfers, timing and incidents are governed by the contractual bank settlement arrangements.

Positions are frozen at closing in accordance with the Book IV calendar, which must be updated. After the prescribed checks and disputes, the authorised institutions execute net settlement instructions. Any residual funds are released or reassigned only after bank confirmation and reconciliation. The former escrow-adjustment windows described in Release 2.1 must be adapted to this bank lock without implicitly altering the contractual duration of a cycle.

7. Canonical Sei EVM Layer and Solvency

The Core, written in pure Solidity, maintains chamber states and clearing positions. At cycle opening, $P_i(0)=C_i$, where C_i represents exclusively bank funds actually locked and verified. For received obligations $R_i(t)$ and issued obligations $E_i(t)$, $P_i(t)=C_i+R_i(t)-E_i(t)$. An obligation is admitted only if $P_i(t)\geq 0$ after posting and all other admissibility and evidence requirements are satisfied. Each balanced obligation preserves the sum of positions; that accounting identity does not, by itself, prove the reality of escrow.

The Core verifies permissions, cycle identifiers, attestation signatures, expiry, uniqueness and prevention of evidence reuse. External data are never trusted merely because they have been transmitted by the oracle.

8. ISO 20022 Messaging and Interfaces

8.a Normative End-of-Cycle Net Settlement Flow

After positions are frozen and prescribed exceptions resolved, Osnias Core publishes a net settlement state for each Osnias-ID and cycle. The node's authorised banking gateway resolves the protected Osnias-ID / IBAN / bank mapping off-chain and generates ISO 20022 orders addressed to the escrow account-servicing institutions. Banks receive these instructions through their agreed interfaces; issuance is based on a payment mandate and conditional escrow-release clauses, without giving the EVM Core or oracle any direct debit authority.

Customer-to-bank orders normally use pain.001 where accepted under the contractual banking arrangement. Banks provide status information, notably through pain.002; posted movements are reconciled using camt.054 and camt.053 (and potentially intraday camt.052). pacs.008 belongs to the interbank flow and is not an order sent directly by the Core to a bank. Message versions, formats, channels and acknowledgements are defined for each bank.

The Bank Escrow Oracle independently checks debits, credits, lock statuses and settlement identifiers. Osnias Core marks a cycle settled only after receiving sufficient bank evidence reconciled for all relevant accounts. Rejections, partial settlement, interbank delays and anomalous operations remain in exception status and do not trigger a fictitious final closing. Residual funds may be released only after confirmed settlement and in accordance with the escrow agreement.

8.b Settlement Security Invariants

Instructions must be unique, idempotent, timestamped, signed by authorised actors and bound to a frozen cycle-state version. Algebraic net balances for each chamber and currency must sum to zero before payments are issued; any discrepancy blocks the batch. No IBAN, identified account balance or sensitive commercial details are published in cleartext on Sei. Loss of bank connectivity, missing confirmation or reconciliation discrepancy suspends closing without early release of funds.

ERP APIs carry obligation requests and acceptances with invoice references and end-to-end identifiers in structured schemas suited to ISO 20022 requirements. Bank payment and reporting messages are distinct: pain.001 for customer-to-bank payment initiation under the contract, pacs.008 for interbank exchange, camt.052 for intraday reporting, camt.053 for statements and camt.054 for entry notifications. None of these messages alone proves the contractual lock of escrow funds.

Sensitive commercial data travel off-chain or as non-reconstructible references. Trading partners' identities are disclosed only to authorised parties, subject to legal audit, anti-money-laundering and supervisory requirements.

8.c — Implementation of the Sei EVM ↔ ISO 20022 ↔ Banking Systems Gateway

The Osnias Bank Gateway is the shared banking interoperability layer of the protocol. It is implemented in Go (Golang), separately from the Osnias Core clearing engine written in standard EVM Solidity. The gateway does not perform clearing: it observes finalised Core states, verifies bank escrow, carries authorised settlement instructions and reconciles bank confirmations. The Core remains portable across compatible EVM chains.

Target software architecture. Sei Connector (Go, EVM JSON-RPC, events and finality checks); ISO 20022 Engine (Go, XML/XSD, bank-specific variants, validation and end-to-end references); Escrow Service (Go, read-only access to balances, movements and blocked amounts); Settlement Service (Go, mandated net settlement orders, status tracking and duplicate-prevention controls); Bank Adapters (REST/HTTPS APIs with mTLS and, where supported, EBICS 3.0 TS); Reconciliation & Audit (Go, matching and audit trail). PostgreSQL stores off-chain transactional states and verifiable audit records. Sensitive signing keys and certificates are protected using HSM/KMS where required by the banking partner.

Escrow Service and Settlement Service are separated by permissions, certificates and keys. The former has read-only and audit access; the latter may submit only net settlement instructions expressly authorised under the applicable bank mandates and escrow agreements. No identifying IBAN data or bank statements are published in clear text on Sei.

The gateway processes pain.001 (payment initiation), pain.002 (payment status), camt.052 (intraday reporting), camt.053 (statement) and camt.054 (notification), using the message versions actually supported by each bank. Evidence that funds are legally and operationally locked must originate from, and be authenticated by, the account-servicing bank; it is not automatically supplied by a generic ISO 20022 message. Evidence is time-stamped, linked to the Osnias-ID and cycle, and verified before admission.

Each net settlement instruction carries a stable, unique reference per bank, cycle and instruction. If confirmation is missing, the gateway retains a PENDING_RECONCILIATION state and queries the bank; it does not automatically issue a replacement order with a new reference. A positive pain.002 status is not, by itself, proof of final settlement. Reconciliation against booked transactions and bank confirmation remains mandatory.

The proof of concept uses a simulated-bank adapter for 1W cycles, followed by a real-bank adapter under contract. Production controls cover on-demand read availability, evidence freshness, connectivity failures, duplicate payments, privilege separation, key management, recovery procedures and independent security audit.

9. Node Infrastructure

Operating a node is a software, contractual and compliance function. The node manages client identity, admission records, ERP integration and coordination with banks. It hosts no EURC escrow smart contract and relays no Ethereum Controller. Its regulatory qualification depends on the services actually performed, including payments, account information, payment initiation, clearing and any custody activities.

10. Bank Escrow Oracle and Other Controls

10.a Interfaces

Service	Verified Data
EscrowRegistry	Authorised Osnias-ID / IBAN / cycle / bank mapping
BankAdapter	Read-only banking API access, bank by bank
EscrowVerifier	Bank balance, locked amount, lock status and expiry
ProofEngine	Authenticity, timestamp, signature, integrity and anti-replay
CoreOracle	Minimal attestation verifiable by the EVM smart contract

10.b Minimum Attestation

Each response must establish the account concerned, currency, amount actually locked, lock status and release conditions, validity period, unique evidence reference, time of observation and the bank's signature or authentication mechanism. Debit and credit movements must be auditable. A balance-only response never activates collateral.

10.c Fail-Closed Policy

Missing, stale, inconsistent, invalidated or non-authoritative banking evidence triggers an EXCEPTION or STALE status and blocks the corresponding new admissions. A time oracle remains

necessary for cycle windows; solvency checks rely on Sei and bank attestations, without an Ethereum/Sei comparator.

11. Financial Effects and Auditability

Netting can reduce gross settlement flows but does not guarantee a fixed reduction rate. Individual escrow accounts provide bounded cover; the bank enforces the lock but does not guarantee settlement in every circumstance. Each operation is reconciled with its Osnias-ID, cycle, bank attestation and observed settlement transactions. The audit trail must capture freshness checks, discrepancies, permissions and incidents.

12. Regulatory Scope and Responsibilities

Removing crypto-assets from settlement and collateral may place nodes outside the scope of crypto-asset service provision where they provide no such services. It does not automatically disapply rules governing payments, account information, clearing, financial market infrastructures, AML/CFT or data protection. The specific arrangements of nodes, escrow banks and settlement finality must be submitted for preliminary assessment by the competent authorities. This document is neither a legal opinion nor an authorisation.

13. Security and Changes from Release 2.1

Removed	Retained / Added
Ethereum and cross-chain Controllers	Single Solidity Core on Sei EVM
Stablecoin / EURC escrow	Individual, non-pooled bank escrow
Ethereum collateral and cross-chain reconciliation oracles	Bank Escrow Oracle with authenticated attestations
Bridges and on-chain value transfers	Net bank settlement
Business logic dependent on a secondary chain	External adapters and EVM portability

Removing components reduces attack surfaces but transfers a critical portion of risk to the banking interface and evidence chain. Audit priorities include access rights, revocation, attestation replay, loss of synchronisation of the bank lock, banking unavailability, protection of IBAN/Osnias-ID mappings, closing permissions and governance of Core upgrades.

14. Conclusion

Osnias Clearing retains a single canonical clearing layer on Sei EVM, developed in pure, portable Solidity. Settlement money remains in the banking system and is effectively locked in an escrow account dedicated to each Osnias-ID throughout the cycle. The oracle verifies, the Core clears and authorised institutions settle. No Ethereum Controller, clearing token, bridge or cross-chain

reconciliation oracle remains in the target architecture. Production deployment depends on securing contractual banking services that guarantee verifiable locking, on-demand access and operational settlement finality. At maturity, each account-servicing bank receives ISO 20022 instructions corresponding to the net positions of its escrow accounts, executes the authorised settlements and provides verifiable confirmations to the Osnias reconciliation process.

Annex A — Matters Requiring Contractual Agreement and Validation

• Legal character of the escrow account and enforceability of the lock, including during insolvency proceedings.

• Signed bank evidence and on-demand API queries, availability, freshness and revocation mechanism.

• Precise conditions permitting only end-of-cycle settlement debits.

• Alignment of Book IV phases and dates with an effective lock throughout the active period.

• Regulatory classification of each operator, the bank and payment services.

• Confidentiality strategy for entries on a public EVM chain and protection of personal data.

• Mandate to transmit ISO 20022 orders to account-servicing banks, authorisations, approval workflow and conditional release solely for net settlement.